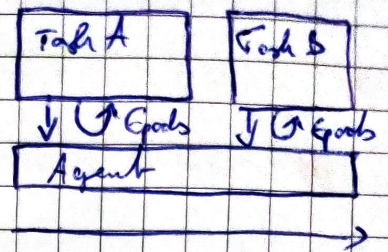


Continual Learning

Learning Strategies

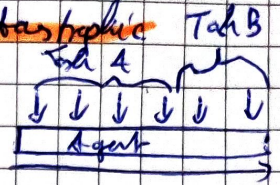
→ Batch Learning :

- Fixed set of tasks
- All training data is available
- If trained with new tasks, prone to catastrophic forgetting or interference



→ Continual Learning :

- Dynamic number of tasks
- No distinction between training and test phases
- Learn new tasks without interfering with existing knowledge
- No access to previously seen training samples



Continual Learning :

- Ability to continually acquire, fine-tune, and transfer new knowledge and skills over time (lifelong learning)
- Overcome catastrophic forgetting when learning from changing input distributions
- No retraining from scratch and no re-access to previously seen data
- Conserved computational and memory resources

Catastrophic Forgetting

- Training a model with new information interferes with previously learned knowledge
- Performance degradation (interference) all the way to a complete overwriting (forgetting) of old knowledge with new one
- Catastrophic forgetting affects all connectionist models

Stability - Plasticity Dilemma

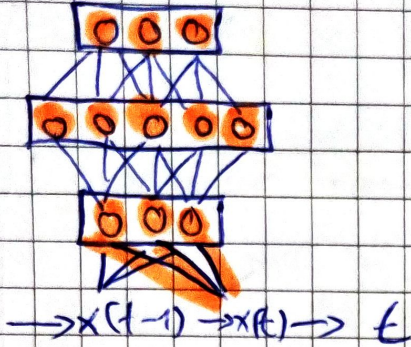
- Stability :
 - Preserve and consolidate structured knowledge
 - Learn in a slow, sustainable manner
 - Generalization
- Plasticity :
 - Adapt quickly to changes in the environment
 - Efficiently learn from novel sensory input
 - Specialization

CL Strategies

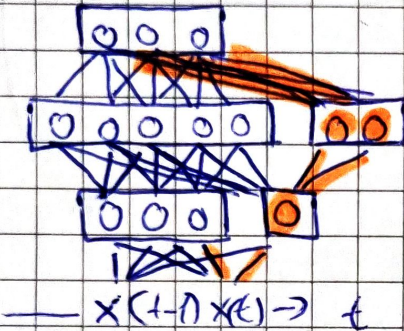
- ⇒ **Regularization**:
 - Constraints on the update of neural weights by means of additional loss terms
 - Habituation of parameters important for previously seen data
 - Fixed model capacity
 - Eg: Elastic Weight Consolidation
- ⇒ **Dynamic Arch:**
 - Apply modular changes to architectural properties in response to novel input
 - Introduce task-specific parameters (eg: add neurons)
 - Increasing model capacity and computational cost with further training
 - Eg: Progressive NN
- ⇒ **Rehearsal / Replay**:
 - Rehearsal: explicitly stored training samples
 - Periodical retraining on a subset of previously seen data
 - Pseudo-rehearsal: Artificial samples are generated from data distribution of prior tasks based on representation learning
 - Generative-Replay: Latent representation of the input
 - Memory requirements and computational cost grows with #tasks
 - Eg: iCaRL
- ↳ **Memory Replay**:
 - Memory reactivation and replay for memory consolidation
 - Cortico-hippocampal interaction
 - during sleep and awake phase
 - disrupting slow wave sleep impairs long-term memory consolidation
 - learn also in the absence of sensory input
- ⇒ **Hybrid Approaches**:
 - Combination of regularization, structural changes and memory replay
 - Dual-memory systems, eg: Growing-Dual-Memory

L13

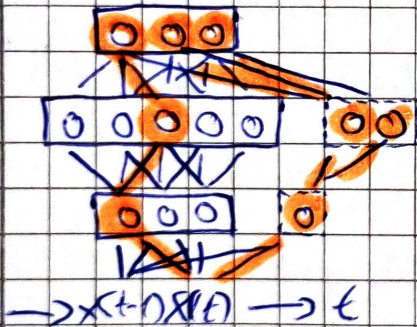
CL strategies for NN



a) Retraining with Regularization



b) Training with network expansion

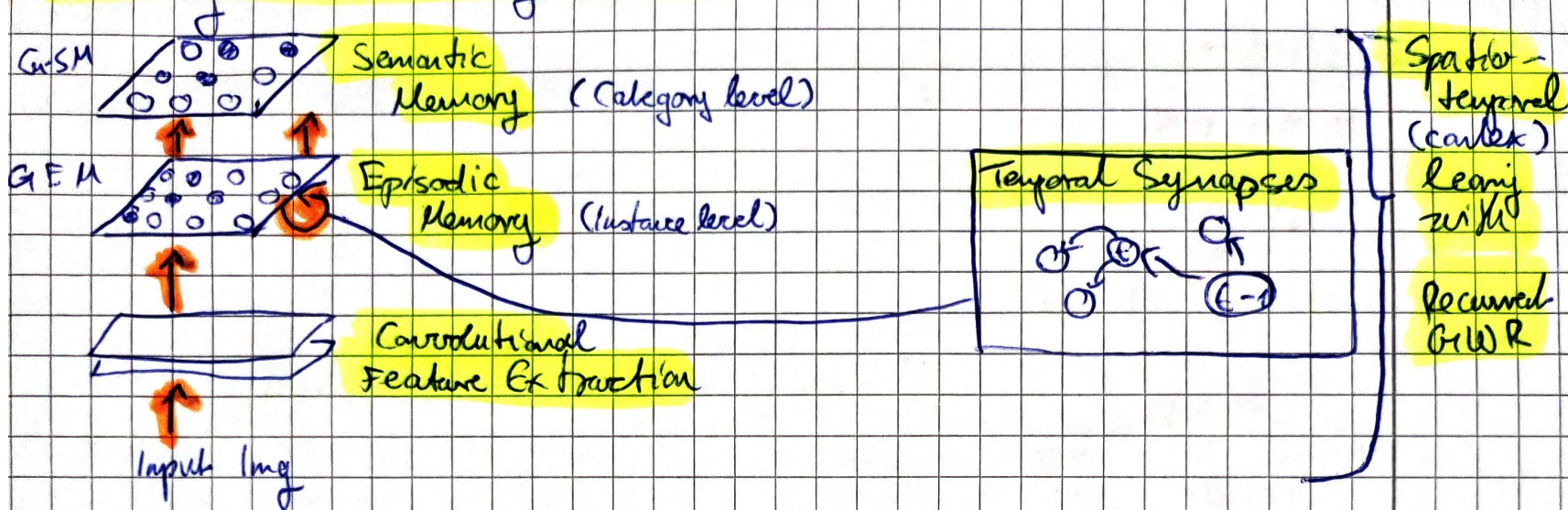


c) Selective Network retraining and expansion

Evaluating CL

- Average Accuracy: $A_k = \frac{1}{k} \sum_{j=1}^k a_{kj}$
- Forgetting: $f_{j,k} = \max_{m \in \{1, \dots, k-1\}} (a_{mj}) - a_{kj}$

Growing Dual-Memory Networks (GDM)



→ Grow - When Required (GWR)

→ Dynamic variant of SOM

→ Graph structure of neurons that learns to represent a dataset

→ Start with few neurons

→ "Grows" by creating new neurons for represent novel input

→ GWR can cope with CL tasks as new neurons can be created to allocate novel knowledge

→ Large GWRs expensive $\Rightarrow$ prune unused neurons

→ Catastrophic interference still possible $\Rightarrow$ protect knowledge

QWR Alg

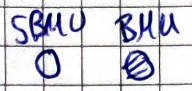
I) $A \leftarrow$ Set of 2 neurons

II) $E \leftarrow$ Set of connections $\emptyset$

III) At each iteration t , receive an input vector $x(t)$

IV) Select BMU & SBMU

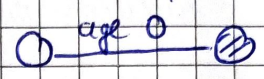
$b = \arg \min_{j \in A} \|x(t) - w_j\|^2, \quad s = \arg \min_{j \in A \setminus \{b\}} \|x(t) - w_j\|^2$



w_j : Out
input

V) IF there is no connection between b and s , create it:

$E = E \cup \{(b, s)\}$



VI) Set the age of this connection to 0 input

VII) Calculate current activation:

// how well it fits the input

$\alpha(t) = \exp(-\|x(t) - w_b\|^2)$

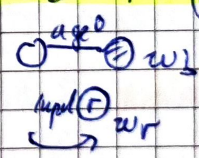
Case 1:

not fit well not firing often

VIII) if ($\alpha(t) < \alpha_{th}$) and ($h_b \leq h_{th}$) then

h_b = firing counter!

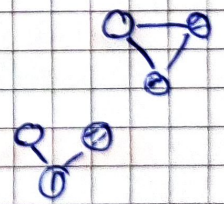
~~IX)~~ Add a new neuron r : $A = A \cup \{r\}$



~~X)~~ Create new weight vector $w_r = 0.5(w_b + x(t))$

~~XI)~~ Connect the new node r with b & s : $E = E \cup \{(r, b), (r, s)\}$

~~XII)~~ Remove the connection between b & s : $E = E \setminus \{(b, s)\}$



Case 2: BMU fits the input well // update BMU and its neighbors

~~XIII)~~ Update BMU and its neighbors: $\Delta w_b = \epsilon_b \cdot d_b \cdot (x(t) - w_b)$
 $\Delta w_i = \epsilon_h \cdot h_i \cdot (x(t) - w_i)$

~~XIV)~~ Increment the age of all edges connected to b : $age(b, i) = age(b, i) + 1$, with $i \neq s$

~~XV)~~ Reduce the firing counters of the BMU & neighbors: $\Delta h_b = \tau_b \cdot K \cdot (1 - h_b) - \tau_b$
 $\Delta h_i = \tau_h \cdot K \cdot (1 - h_i) - \tau_h$

~~XVI)~~ Remove each edge (i, j) with

$age(i, j) > \mu_{max}$ and remove isolated nodes

• GWR

- Classification: How GWR perform classification? → Associative GWR
 - For each neuron, keep a list of the classes of all inputs for which the neuron was the BMU
 - When a GWR classifies a novel input, determine the BMU and determine the class with the most entries in the neuron list
- Expandable / Shrinkable Networks
- Hebbian-like structural plasticity
- Input-driven self-organization
- Neurogenesis: Neuron activation, habituation

• Continuous Language Learning (CLL)

- Applications & Tasks:
 - Question Answering
 - Text Classification / Sentiment Analysis
 - Machine Translation

→ NLP: Natural Language Processing

→ Challenges: Word / Sentence Representations

- State of the art language embeddings are usually generated from pretraining on massive cross domain, general purpose corpora

(-) Limitation: Decreasing performance on domain-specific downstream tasks w.r.t. changes of vocabulary in time and domain

- Current Solution: Meta-learning of prior knowledge in order to generate improved new domain embedding

→ Challenges: Fine-Tuning Language Models

- Transformer-based language models are pretrained on large-scale corpora and then have to be fine-tuned/tailored for the task at hand

(-) Limitation: Knowledge about all tasks must be present prior to designing the model

- Current Solution:

1) all tasks viewed through the lens of QA ^{ask}

2) not only sequences, but also info about domain and task are used as input for general-purpose models

CLL Future Direction (Bio-Inspired)

- learning high-level language concepts and compositional structures (compositional / systematic generalization)
- storing semantic concepts from previously learned tasks in a compact fashion (knowledge distillation)
- recombining semantic concepts when being exposed to novel tasks (abductive reasoning)

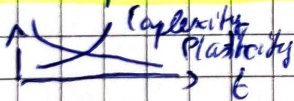
State of the Art (SotA)

- transformer-based language models, e.g. Bidirectional Encoder Representations from Transformers (BERT)

Step 1: Unsupervised Pretraining, Step 2: Supervised Finetuning

Related Paradigms

a) Developmental & Curriculum Learning



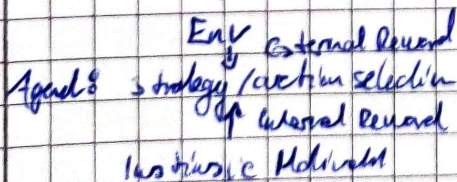
- Critical periods of learning
- early experience are the most influential
- curriculum of increasingly complex tasks

b) Multi-Task Transfer Learning



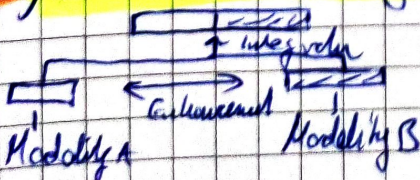
- Apply knowledge in one domain to a novel task
- Positive and negative transfer
- Forward and backward transfer

c) Curiosity and Intrinsic Motivation



- Select strategies that maximizes reward
- Empirical process of exploration
- self-generation of a learning curriculum

d) Crossmodal Learning



- Multisensory integration and crossmodal enhancement
- Dynamic process across a lifetime
- Address sensory uncertainty and conflict resolution

Summary

- CL is the ability to incrementally learn over time without forgetting
 - Stability - Plasticity Dilemma
 - Bio & Hebbian learning & Complementary Systems Theory
- Overcome catastrophic forgetting in NN
- Continual Object Recognition
 - CL on sequences with Grammar GWR
 - GDM & hybrid model with memory replay
- NLP: CL and Challenges
- Current Models: far from providing flexibility, robustness & scalability
- Basis for modeling higher-level cognitive functions in AI